

SpaceLoop: Risk, Legal Liability & Privacy Architecture

Authoritative Operational & Legal Governance for Property Damage, Emergency Hazards & Location Privacy

Subject: Legal Liability, Emergency Protocols & Location Privacy	Governing Acts: Easements Act 1882 / IT Act 2000 / DPDP 2023	Audience: Hosts, Renters, Legal Reviewers & Judges
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Question 1: If anything goes wrong at the property during the booking and it damages the property, who is responsible? The owner? The renter? Or the platform?

Core Operating Principle: To minimize systemic harm and administrative friction, responsibility is allocated strictly to the party best positioned to prevent the hazard (the least-cost avoider), supported by objective visual evidence and legally binding contracts.

1. The Renter's Responsibility (Renter-Induced Damage):

- **Direct Civil & Financial Liability:** If the renter or their attendees cause physical damage, breakage, stains, or equipment loss, the renter bears sole responsibility.
- **Contractual Enforcement (Section 52, Indian Easements Act, 1882):** Every reservation generates an enforceable **AI Micro-Lease** granting a temporary revocable license. It includes an explicit **Indemnification Clause** under which the renter assumes legal liability for restoring the space to its pre-booking baseline and indemnifies the host against misuse.
- **Objective Evidence via AI Visual Diff:** The platform compares the 15-second check-in walkthrough baseline against the departure scan. If damage is detected, the host's claim is substantiated by objective mathematical visual comparison, eliminating 'he-said-she-said' disputes.
- **Recovery Mechanism:** The Rs. 100 UPI micro-escrow is withheld immediately. For major damages, the host has direct legal recourse against the renter backed by their verified **DigiLocker Aadhaar token hash** and institutional college identity.

2. The Owner's Responsibility (Structural & Pre-Existing Defects):

- **Premises Inherent Hazards:** If damage occurs due to pre-existing structural decay, faulty plumbing, roof leakage, or electrical wiring inherent to the building, the **owner is solely responsible**.
- **Renter Protection:** If a structural failure interrupts the session through no fault of the student, the platform voids the booking, issues an immediate 100% refund for unused hours, and releases the Rs. 100 escrow deposit in full.

3. The Platform's Role (SpaceLoop as Intermediary):

- **Statutory Safe Harbor (Section 79, Information Technology Act, 2000):** SpaceLoop is a digital marketplace and technological enabler. It does not own, lease, or physically insure the real estate.
- **Preserving Economic Accessibility:** If the platform absorbed open-ended physical tort liability for private properties, it would be forced to levy high insurance surcharges, inflating prices from Rs. 50/hr to over Rs. 300/hr and pricing out students. Instead, SpaceLoop maximizes societal utility by providing cryptographic presence proof, verified identity tokens, and automated escrow arbitration.

Incident Scenario	Primary Liable Party	Evidentiary Basis	Financial Remedy
Spilled drink on carpet / broken chair	Renter	AI exit visual diff shows new stain/breakage	Rs. 100 escrow forfeited; excess billed via KYC identity
Plumbing pipe bursts in wall	Host / Property Owner	Structural failure unrelated to renter activity	Host absorbs repair; renter receives full refund + escrow
Missing items / stolen monitor	Renter	Item present in check-in photo, missing at exit	Police report filed with Aadhaar & college records
Platform technology failure (app glitch)	SpaceLoop	Server log timestamps and telemetry outage	Full booking refund + reservation credits issued

Question 2: If somebody doesn’t want to share their location, what can we do?

Core Operating Principle: Balancing individual data privacy against the collective need for fraud prevention. Denying location permissions should never result in outright exclusion. Instead, the platform uses a **graceful degradation protocol**, offering three alternative, non-satellite physical presence verification paths.

Fallback 1: Two-Way Caretaker / Host 4-Digit PIN Handshake (Zero-Data Alternative)

When GPS is denied, the student app displays a dynamic, time-sensitive **4-digit Arrival PIN** (e.g., 4821). The student conveys this code in-person to the property caretaker or host. The caretaker confirms the code on their own portal, or provides the matching mechanical keybox combination. Physical human interaction replaces satellite tracking with zero personal data transmitted.

Fallback 2: Local In-Room Wi-Fi Network Footprint Match

If the student connects their smartphone or laptop to the property's local Wi-Fi router, the browser verifies the network's public IP gateway or BSSID against the verified router profile logged by the host during Discom onboarding. Connecting through the room's physical router provides unforgeable proof of co-location without accessing device GPS sensors.

Fallback 3: Live In-App Camera Walkthrough (Anti-Spoofing Stream)

If both GPS and Wi-Fi matching are unavailable, the check-in console disables gallery file uploads and activates a live camera stream. The user must scan the laminated door QR pass in real time and capture a 5-second video pan of the interior doorway. Server-embedded optical watermarks confirm physical presence at the doorway.

Verification Method	Data Shared by User	Fraud Resistance	User Privacy Level
GPS Geofencing (<50m)	Coordinates at check-in/out instant only	High (Satellite-verified)	Standard (No continuous tracking)
Caretaker 4-Digit PIN	None (Offline verbal exchange)	Very High (Physical human handshake)	Maximum (Zero sensor telemetry)
Local Wi-Fi Match	Router public IP gateway footprint	High (Physical router connection)	High (Zero geographic coordinates)
Live Camera Stream	5-second real-time camera feed	High (Anti-spoofing live watermark)	High (Interior room visual only)

Question 3: What if the renter is present in the property and a fire breaks out? Who will be responsible?

Core Operating Principle: When a catastrophic hazard occurs, human safety supersedes contractual preservation. Subsequent financial and legal liability is governed strictly by **causation and negligence**—holding accountable the party who introduced the hazard or failed standard safety duties.

1. Scenario A: Renter Negligence (Renter is Liable)

- **Prohibited Activities:** The AI Micro-Lease expressly prohibits open flames, smoking, cooking stoves, candles, and incense. If a fire starts from any prohibited activity, the renter is **solely liable** under both civil and criminal law.
- **Electrical Overload:** Connecting unapproved high-power appliances (e.g. heating coils, space heaters, crypto-mining rigs) that exceed the room's declared circuit capacity (e.g., pulling 30A on a 5A line) constitutes gross negligence by the renter.
- **Enforcement:** The Rs. 100 escrow is forfeited immediately. The host and building insurer can pursue full structural recovery under the signed **Micro-Lease Indemnification Clause**, backed by the renter's verified DigiLocker Aadhaar identity.

2. Scenario B: Premise / Electrical Infrastructure Defect (Host is Liable)

- **Defective Building Wiring:** If forensic investigation (Fire Department / Police report) determines the fire originated from aging internal wall wiring, faulty main circuit breakers, or an unserviced host appliance (AC/fan short-circuit), the **host is solely liable**.
- **Premises Liability:** Under Indian tort law, an owner granting a license is legally obligated to ensure the premises are reasonably safe. The host is liable for any injury to the renter or loss of the renter's personal property (e.g. student laptops, bags).
- **Renter Protection:** SpaceLoop voids the booking immediately, issues a 100% refund, releases the Rs. 100 escrow, and permanently delists the property until municipal fire clearance and Discom electrical fitness certificates are produced.

3. Scenario C: External Force Majeure (No-Fault / Insurance Coverage)

- If a fire spreads from an adjacent building, results from an external municipal grid surge, or is caused by natural lightning, the **Force Majeure Clause** applies. Neither host nor student is penalized.
- The student receives an immediate full refund and escrow release. The host's structural damages are resolved through standard property fire insurance.

4. SpaceLoop's Pre-Emptive Protocols & In-App Emergency Response:

- **Pre-Booking AI Circuit Audits:** The AI Space Inspector flags electrical limits (5A vs 15A outlets) to deter hazardous bookings.
- **Live In-Room Emergency SOS:** The In-Room Console (`/booking/<id>/session`) features a prominent, one-tap Emergency SOS connecting directly to **National Emergency 112** and **Fire Services 101** with pre-formatted address coordinates.
- **Forensic Audit Records:** SpaceLoop provides verified check-in photos, arrival timestamps, and declared usage purposes to fire investigators.

Fire Cause / Origin	Determining Evidence	Liable Party	Legal & Financial Outcome
Candle / cigarette / open flame	Fire forensic report confirms localized flame origin	Renter	Civil damages billed via Micro-Lease; police report filed
High-wattage electrical overload	Renter declared 'study' but plugged in heavy industrial heater	Renter	Indemnification clause enforced; renter absorbs damages
Internal wall wiring short-circuit	Forensic report confirms pre-existing faulty electricals	Host / Property Owner	Host liable; renter gets 100% refund + personal property damages
Fire spreading from adjacent shop	Municipal fire report confirms external third-party origin	Force Majeure (No Fault)	Covered by building fire insurance; student refunded in full

Summary: SpaceLoop's framework aligns legal clarity, objective sensor data, and human privacy to deliver frictionless micro-spaces.

Target Event: Hack2Ignite 2026